INCH-POUND

AN849 Rev 13
w/ AMENDMENT 1
1 April 2025
SUPERSEDING
AN849 Rev 13
10 March 2015

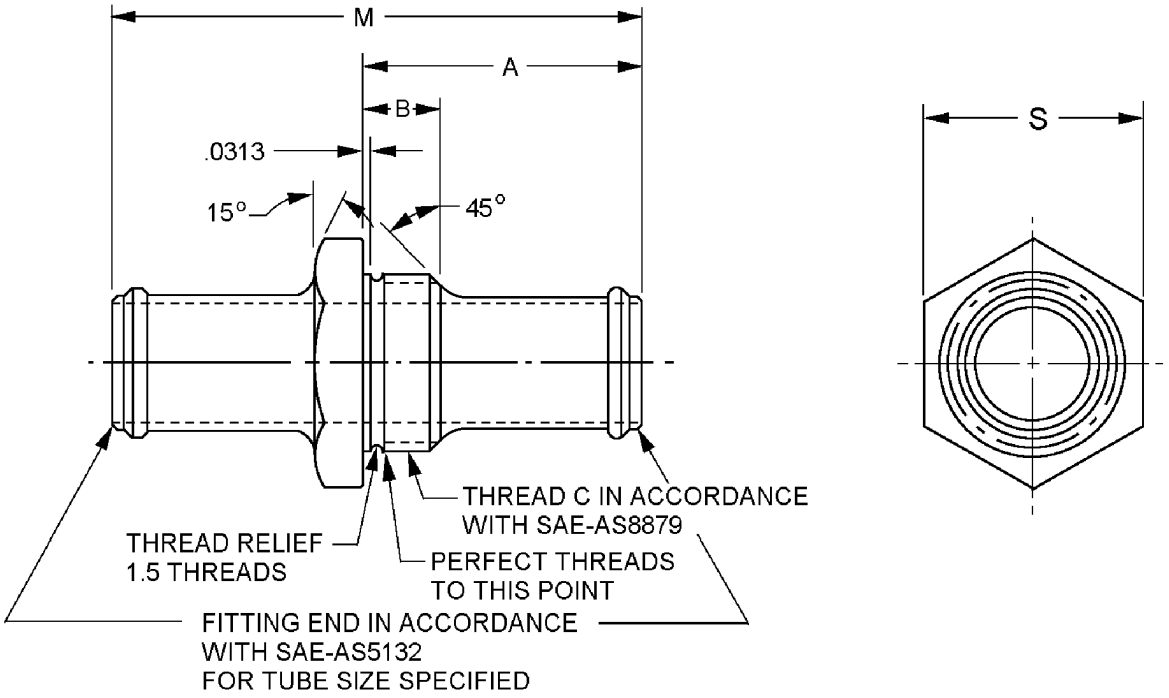
DETAIL SPECIFICATION SHEET

NIPPLE, HOSE

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5189.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843.



Inch	mm
.0313	0.795

FIGURE 1. Nipple dimensions and configuration.

AMSC N/A
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FSC 4730

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Dash number	Hose ID inches (mm)	A inches (mm)	B inches (mm)	Thread C Ref SAE-AS8879	M inches (mm)
4	.250 (6.35)	1.938 (49.23)	.438 (11.13)	.4375-20UNJF-3A	3.625 (92.08)
6	.375 (9.53)	1.969 (50.01)	.469 (11.91)	.5625-18UNJF-3A	3.719 (94.46)
8	.500 (12.70)	2.000 (50.80)	.500 (12.70)	.7500-16UNJF-3A	3.781 (96.04)
10	.675 (17.15)	2.031 (51.59)	.531 (13.49)	1.875 -12UNJ-3A	3.844 (97.64)
12	.750 (19.05)	2.063 (52.40)	.563 (14.30)	1.0625 -12UNJ-3A	3.906 (99.21)
16	1.000 (25.40)	2.063 (52.40)	.563 (14.30)	1.3125-12UNJ-3A	3.906 (99.21)
20	1.250 (31.75)	2.063 (52.40)	.563 (14.30)	1.6250-12UNJ-3A	3.938 (100.03)
24	1.500 (38.10)	2.063 (52.40)	.563 (14.30)	1.8750-12UNJ-3A	4.000 (101.60)

Dash number	S inches (mm)	
4	.688 (17.48)	±.004 (0.10)
6	.813 (20.65)	
8	1.000 (25.40)	
10	1.125 (28.58)	±.005 (0.13)
12	1.375 (34.93)	
16	1.625 (41.28)	±.016 (0.41)
20	1.875 (47.63)	
24	2.125 (53.98)	±.020 (0.51)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges and remove all hanging burrs and slivers.
5. Machined surfaces shall be finished to 125µ-in Ra, forged surfaces shall be 250µ-in Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. Nipple dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Materials and finishes shall be in accordance with SAE-AS4843, see table I for material code.

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TABLE I. Material and finish code letters.

Material and finish code	Material	Protective chemical finish
No code 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/ 4/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 2014-T6. Aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 alloy 2024-T6, or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339 alloy 2024-T851.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 2014-T6. Aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 alloy 2024-T6, or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339 alloy 2024-T851.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
J	Type 304 corrosion resistant steel (CRES) forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
K	Type 316 corrosion resistant steel (CRES) forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
R	Type 321 corrosion resistant steel (CRES) forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
S 3/	Type 347 corrosion resistant steel (CRES) forging or bar in accordance with SAE-QQ-S-763 or SAE-AMS5646	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.

See notes at end of table

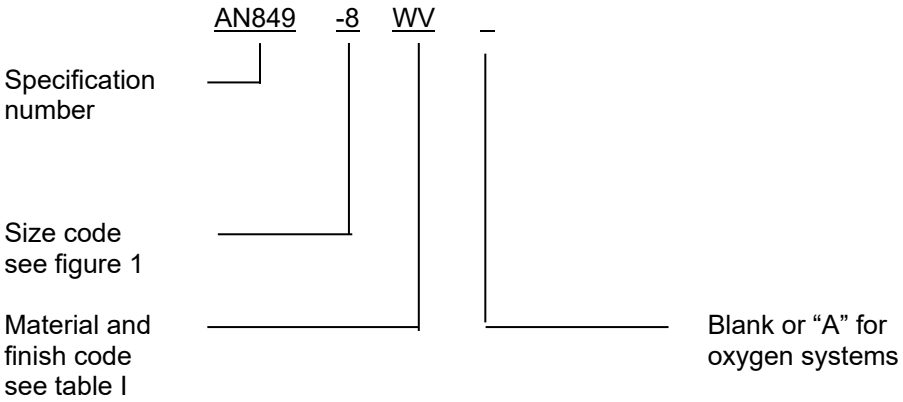
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TABLE I. Material and finish code letters - Continued.

Material and finish code	Material	Protective chemical finish 3/ 4/
T 4/	Titanium in accordance with SAE-AMS4928 alloy 6AI-4V annealed.	Anodize in accordance with SAE-AMS2488, type 2. 5/
TF 4/	Titanium in accordance with SAE-AMS4928 alloy 6AI-4V annealed.	Fluoride phosphate in accordance with SAE-AMS2486. 5/
W 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 7075-T73 or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9 or SAE-AMS4124, alloy 7075-T7351 or 7075-T73.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye brown similar to color 10080 in accordance with SAE-AMS-STD-595, duplex seal in accordance with procurement specification.
WV 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 7075-T73 or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9 or SAE-AMS4124, alloy 7075-T7351 or 7075-T73.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.

1/ Material code was dash on revision 11, changed to agree with SAE-ARP1590.
2/ Aluminum code D is inactive for new design, use code W.
3/ CRES 347 code S is inactive for new design, use code R.
4/ Titanium and cadmium shall not be used in potable water or oxygen systems.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.

Part or Identifying Number (PIN): The PIN consists of the prefix “AN”, specification sheet number, dash number for hose and flared tube size, material finish code letter(s) and blank of “A” for oxygen cleaning requirements. Unassigned PIN’s shall not be used.



PIN examples:

AN849-8W indicates a straight nipple .500 inch (.12.70 mm), aluminum alloy 7075-T73.

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AN849-8WV indicates a straight nipple .500 inch (.12.70 mm), aluminum alloy 7075-T73 finish with high purity aluminum.

AN849-8WA indicates a straight nipple .500 inch (.12.70 mm), aluminum alloy 7075-T73 for use on oxygen systems.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Supersession data:

Due to stress corrosion cracking aluminum alloys 2014 and 2024, “D” designator has been replaced by aluminum alloy 7075 “W” designator. Example: AN849-8D use AN849-8W.

Metal cracking due to high temperatures CRES alloy 347 “S” designator has been replaced by CRES alloy 321 “R” designator. Example: AN849-8S use AN849-8R.

Inactive AN PIN and superseding AN PIN see table II.

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TABLE II. Superseding AN PIN.

Inactive AN PIN	Hose Size	Superseding AN PIN Inactive part
AN849-4	0.250	AN849-4B
AN849-4	0.250	AN849-4BC
AN849-4D	0.250	AN849-4W
AN849-4DV	0.250	AN849-4WV
AN849-4S	0.250	AN849-4R
AN849-6	0.375	AN849-6B
AN849-6	0.375	AN849-6BC
AN849-6D	0.375	AN849-6W
AN849-6DV	0.375	AN849-6WV
AN849-6S	0.375	AN849-6R
AN849-8	0.500	AN849-8B
AN849-8	0.500	AN849-8BC
AN849-8D	0.500	AN849-8W
AN849-8DV	0.500	AN849-8WV
AN849-8S	0.500	AN849-8R
AN849-10	0.625	AN849-10B
AN849-10	0.625	AN849-10BC
AN849-10D	0.625	AN849-10W
AN849-10DV	0.625	AN849-10WV
AN849-10DV	0.625	AN849-10WV
AN849-10S	0.625	AN849-10R
AN849-12	0.750	AN849-12B
AN849-12	0.750	AN849-12BC
AN849-12D	0.750	AN849-12W
AN849-12DV	0.750	AN849-12WV
AN849-12R	0.750	AN849-12S
AN849-16	1.000	AN849-16B
AN849-16	1.000	AN849-16BC
AN849-16D	1.000	AN849-16W
AN849-16DV	1.000	AN849-16WV
AN849-16S	1.000	AN849-16R
AN849-20	1.250	AN849-20B
AN849-20	1.250	AN849-20BC
AN849-20D	1.250	AN849-20W
AN849-20DV	1.250	AN849-20WV
AN849-20S	1.250	AN849-20R
AN849-24	1.500	AN849-24B
AN849-24	1.500	AN849-24BC
AN849-24D	1.500	AN849-24W
AN849-24DV	1.500	AN849-24WV
AN849-24S	1.500	AN849-24R

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Cross reference data: SAE- AS5189 replacement numbers may be substituted for AN849 PINs, see table III.

Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

TABLE III. Cross-reference data. 1/

AN PIN Inactive for new design	Hose Size	Replacement AS PIN	New design
AN849-4	0.250	AS5189B04 2/	
AN849-4D	0.250	AS5189W04	AS5189W04
AN849-4DV	0.250	None	
AN849-4J	0.250	AS5189J04	
AN849-4K	0.250	AS5189K04	
AN849-4R	0.250	AS5189R04	
AN849-4S	0.250	AS5189R04	AS5189R04
AN849-4T	0.250	None	
AN849-4TF	0.250	None	
AN849-4W	0.250	AS5189W04	
AN849-4WV	0.250	None	
AN849-6	0.375	AS5189B06 2/	
AN849-6D	0.375	AS5189W06	AS5189W06
AN849-6DV	0.375	None	
AN849-4J	0.375	AS5189J06	
AN849-4K	0.375	AS5189K06	
AN849-6R	0.375	AS5189R06	
AN849-6S	0.375	AS5189R06	AS5189R06
AN849-4T	0.375	None	
AN849-4TF	0.375	None	
AN849-6W	0.375	AS5189W06	
AN849-6WV	0.375	None	
AN849-8	0.500	AS5189B08	
AN849-8D	0.500	AS5189W08	AS5189W08
AN849-8DV	0.500	None	
AN849-8J	0.500	AS5189J08	
AN849-8K	0.500	AS5189K08	
AN849-8R	0.500	AS5189R08	
AN849-8S	0.500	AS5189R08	AS5189R08
AN849-8T	0.500	None	
AN849-8TF	0.500	None	
AN849-8W	0.500	AS5189W08	
AN849-8WV	0.500	None	

See notes at end of table.

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TABLE III. Cross-reference data - Continued. 1/

AN PIN Inactive for new design	Hose Size	Replacement AS PIN	New design
AN849-10	0.625	AS5189B10 2/	
AN849-10D	0.625	AS5189W10	AS5189W10
AN849-10DV	0.625	None	
AN849-10J	0.625	AS5189J10	
AN849-10K	0.625	AS5189K10	
AN849-10R	0.625	AS5189R10	
AN849-10S	0.625	AS5189R10	AS5189R10
AN849-10T	0.625	None	
AN849-10TF	0.625	None	
AN849-10W	0.625	AS5189W10	
AN849-10WV	0.625	None	
AN849-12	0.750	AS5189B12 2/	
AN849-12D	0.750	AS5189W12	AS5189W12
AN849-12DV	0.750	None	
AN849-12J	0.750	AS5189J12	
AN849-12K	0.750	AS5189K12	
AN849-12R	0.750	AS5189R12	
AN849-12S	0.750	AS5189R12	AS5189R12
AN849-12T	0.750	None	
AN849-12TF	0.750	None	
AN849-12W	0.750	AS5189W12	
AN849-12WV	0.750	None	
AN849-16	1.000	AN5189B16 2/	
AN849-16DV	1.000	AS5189W16	AS5189W16
AN849-16D	1.000	None	
AN849-16J	1.000	AS5189J16	
AN849-16K	1.000	AS5189K16	
AN849-16R	1.000	AS5189R16	
AN849-16S	1.000	AS5189R16	AS5189R16
AN849-16T	1.000	None	
AN849-16TF	1.000	None	
AN849-16W	1.000	AS5189W16	
AN849-16WV	1.000	None	
AN849-20	1.250	AS5189B20 2/	
AN849-20D	1.250	AS5189W20	AS5189W20
AN849-20DV	1.250	None	
AN849-20J	1.250	AS5189J20	
AN849-20K	1.250	AS5189K20	
AN849-20R	1.250	AS5189R20	
AN849-20S	1.250	AS5189R20	AS5189R20
AN849-20T	1.250	None	
AN849-20TA	1.250	None	
AN849-20W	1.250	AS5189W20	
AN849-20WV	1.250	None	

See notes at end of table.

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TABLE III. Cross-reference data - Continued. 1/

AN PIN Inactive for new design	Hose Size	Replacement AS PIN	New design
AN849-24	1.500	AS5189B24 2/	
AN849-24D	1.500	AS5189W24	AS5189W24
AN849-24DV	1.500	None	
AN849-24J	1.500	AS5189J24	
AN849-24K	1.500	AS5189K24	
AN849-24R	1.500	AS5189R24	
AN849-24S	1.500	AS5189R24	AS5189R24
AN849-24T	1.500	None	
AN849-24TF	1.500	None	
AN849-24W	1.500	AS5189W24	

1/ For new design use material designator R and W.
2/ SAE part may be unplated or cadmium plated.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4843, this document references the following:

MIL-PRF-8625
MIL-DTL-83488
ASME B46.1
ASTM B124/B124M
ASTM B138/B138M
SAE-AMS-STD-595
SAE-AMS-QQ-A-225/6
SAE-AMS-QQ-A-225/9
SAE-AMS-QQ-A-367
SAE-AMS-QQ-P-416
SAE-AMS-QQ-S-763
SAE-AMS2472
SAE-AMS2486
SAE-AMS2488
SAE-AMS2700
SAE-AMS4124
SAE-AMS4339
SAE-AMS4614
SAE-AMS4928
SAE-AMS5639
SAE-AMS5645
SAE-AMS5646
SAE-AMS5648
SAE-ARP1590
SAE-AS5132
SAE-AS5189
SAE-AS8879

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CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-2025-036)

Review activities:
Navy - MC, SA

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